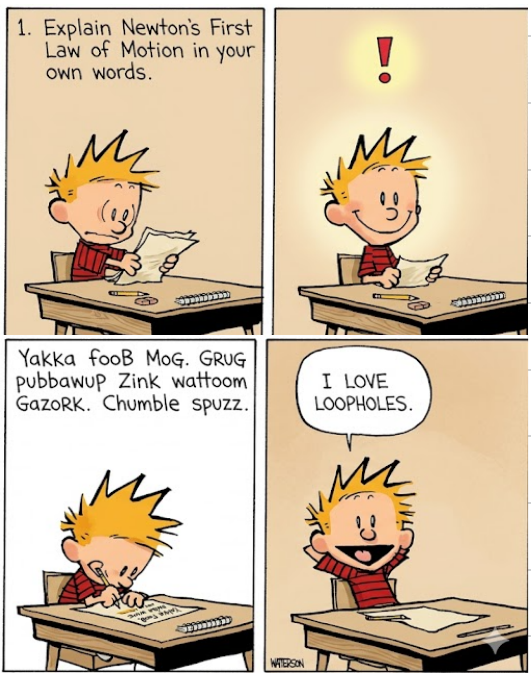
Session 1

Neural Network - 1

March 30, 2026



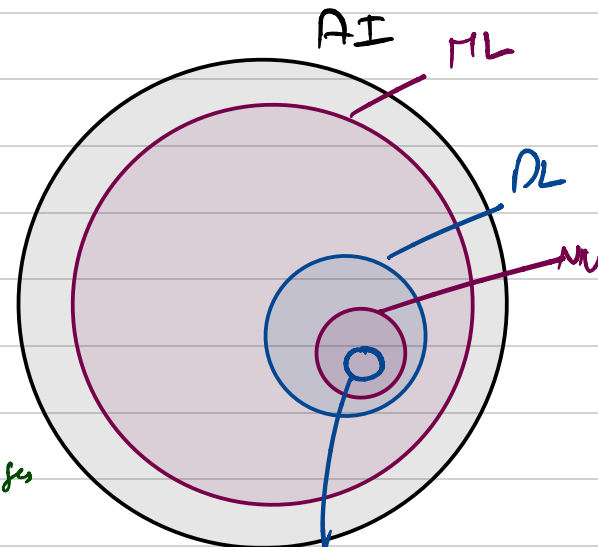


AGENDA

- Why Neural Networks? - Limitations of Classical ML
- Real-World Applications of Deep Learning
- The Single Neuron Model
- Building Neural Network Architecture

What we've done So Far??

→ Machine Learning
→ Deep Learning
  → Basic
  → CNN
  → NLP



Neural Network →

Convolutional NN → Images

Recurrent NN → texts

GANs →

GPT ← Transformers

Multimodal ← transf

$f_1, f_2, \ldots f_n$	y



$F_1, F_2, \ldots$ → Spatial prop

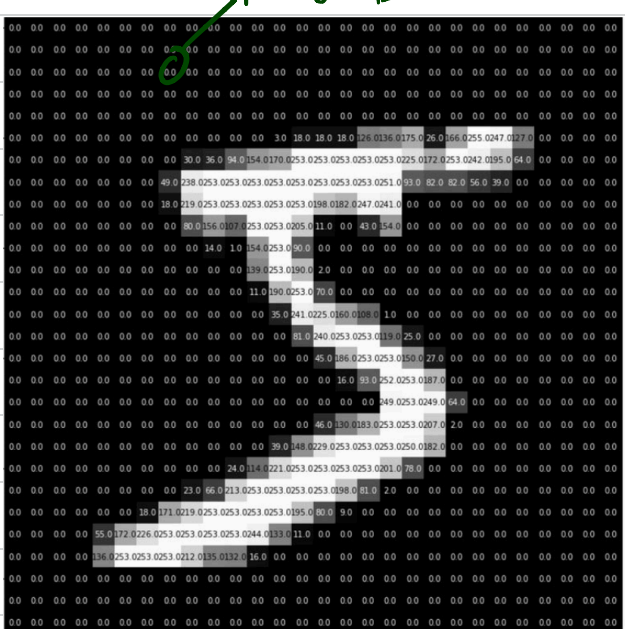
$F_1, F_2 - -$

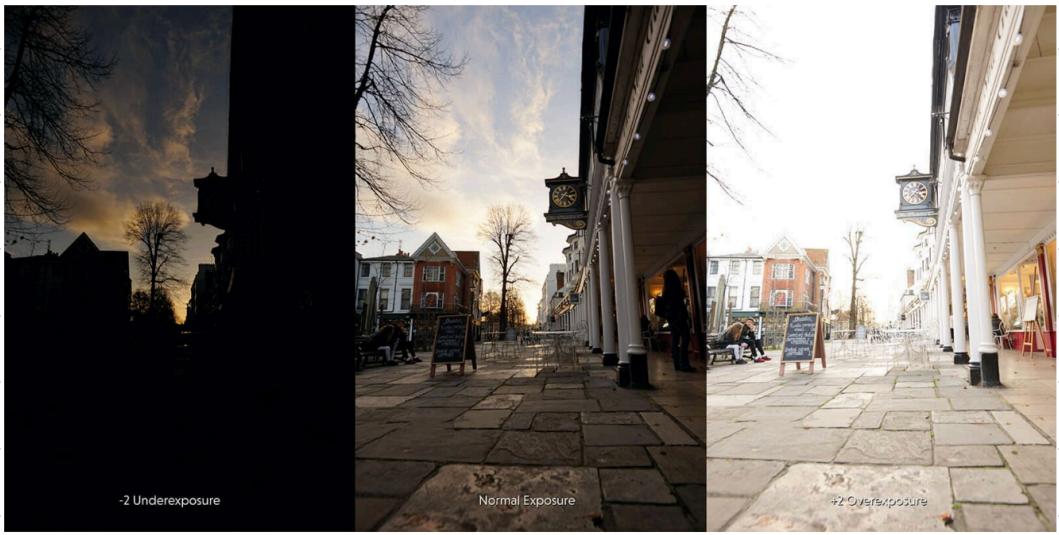


R-G-B

0 – 255

0 — 1









Real examples that break simple word counting

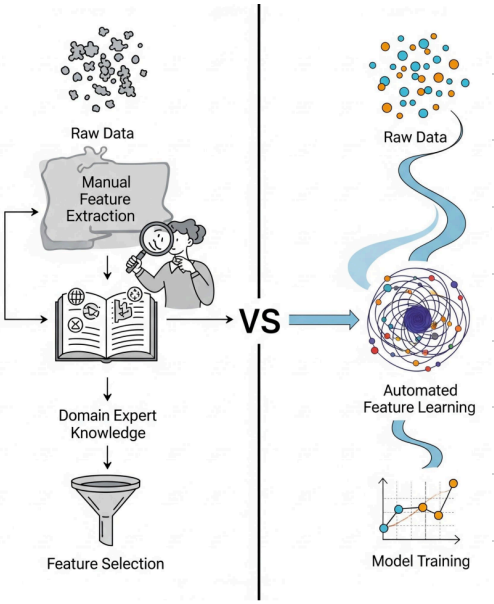
- "The food was not great, but not terrible" → mixed, needs context
- "I love cats. I hate this movie." → sentiment depends on what you're talking about
- "It was good. Really good. Totally good." → repetition for emphasis
- "Best worst movie ever" → contradictory words
- "Can't wait to never see this again" → negation + sarcasm

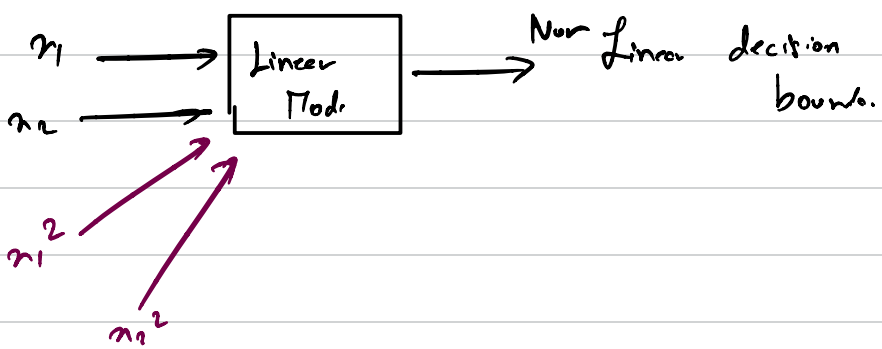
Same word, different meaning

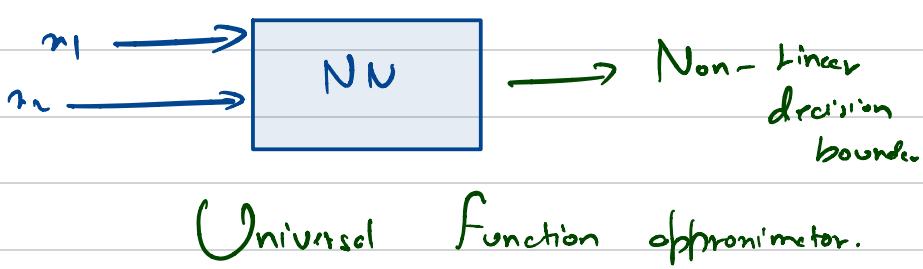
- “That was sick” → positive (slang)
- “I feel sick” → negative (literal)

Sarcasm breaks keyword logic

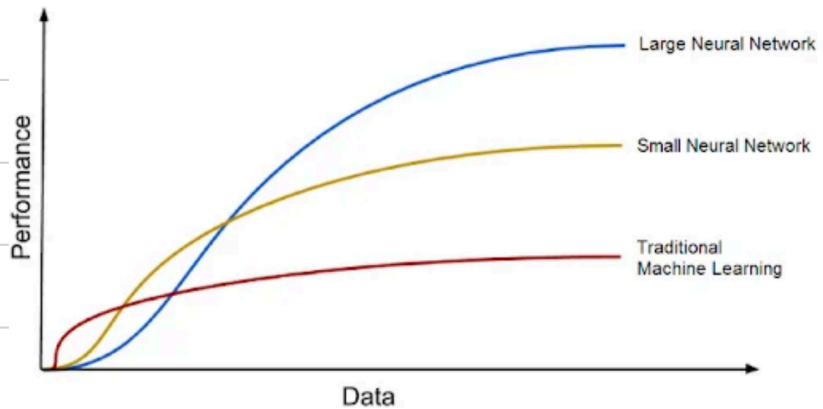
- “Yeah, great job…” (tone flips meaning)







Universal Function approximator.



Tabular dataset -> GBDT based algos.

Image / Text -> NN

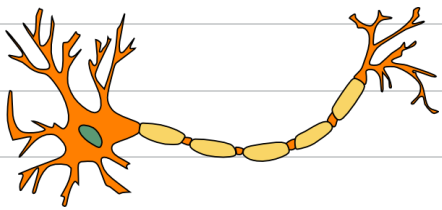
Real World applications DL:

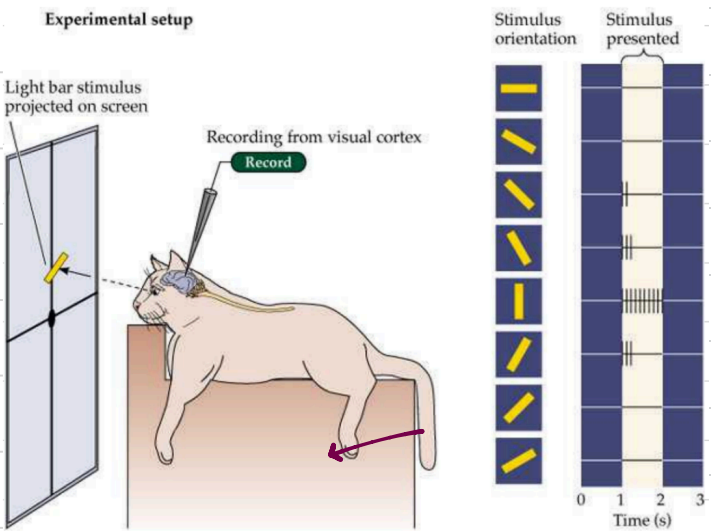
1. OCR
2. Recommendations
3. ADAS

Auto Encoder

	f_1	f_2	f_3	f_4	f_5	f_6	y
x_1							
x_2							
x_3							
x_4							
x_5							
x_6							
x_7							
x_8							

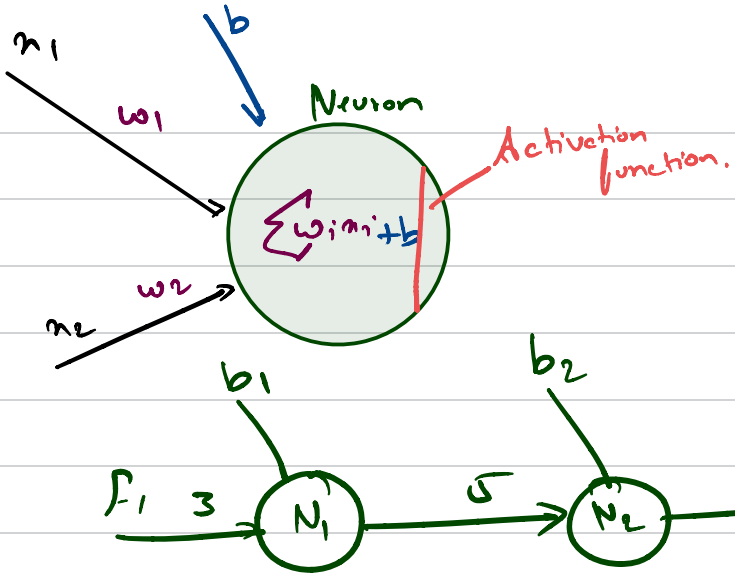




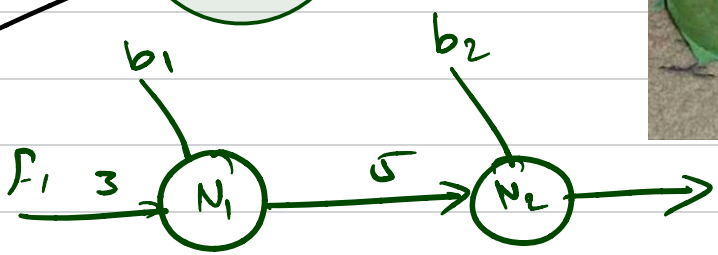


Neurons that Fire together wire together.

n_1 & n_2 → Feeton







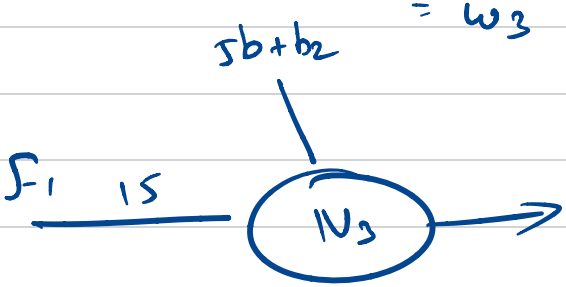
$$N_1 = 3 \times F_1 + b_1$$

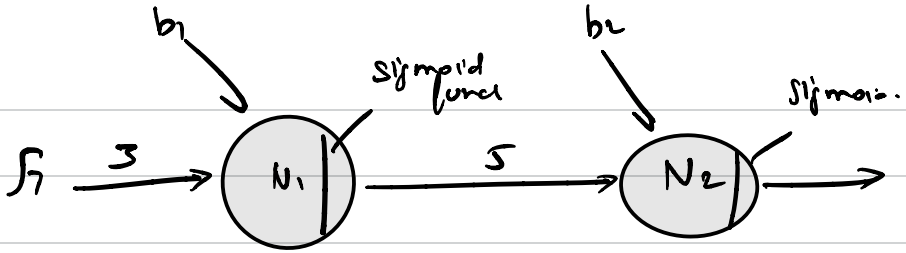
N_2

$$5 \times (3 \times F_1 + b_1) + b_2$$

$$N_2 = 15 F_1 + 5 b_1 + b_2$$

$$= \underbrace{15 F_1}_{= w_3} + \underbrace{(5 b_1 + b_2)}_{b_3}$$





$$N_1 = \sigma(3 \times f_1 + b_1)$$

$$N_2 = \sigma(5 \times N_1 + b_2)$$

$$\Rightarrow \sigma(5 \times \sigma(3 \times f_1 + b_1) + b_2)$$

Activation function allow you to create non-linear combination of features

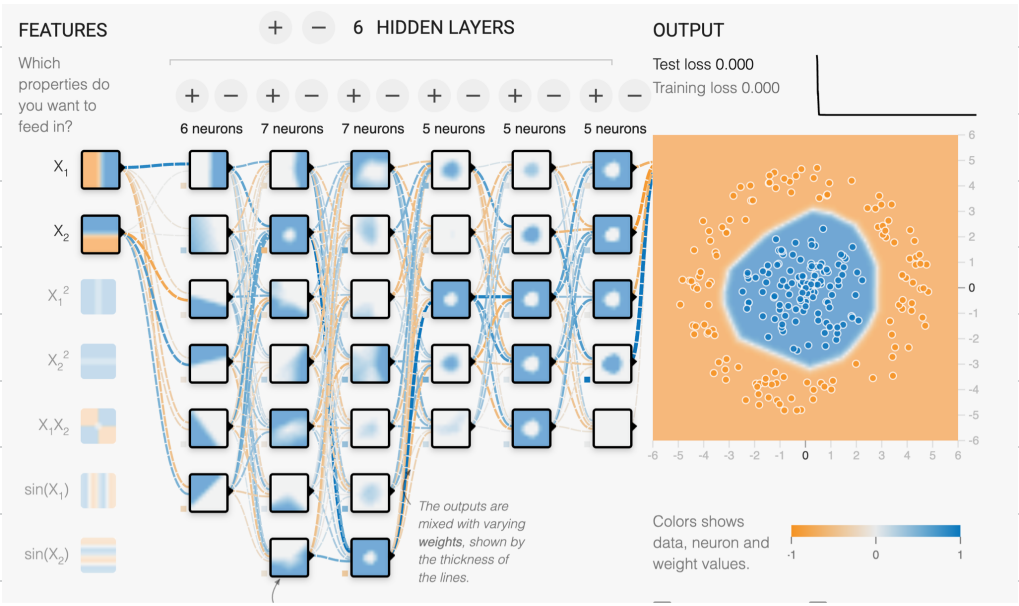
$a - 5 \rightarrow \sigma(5)$

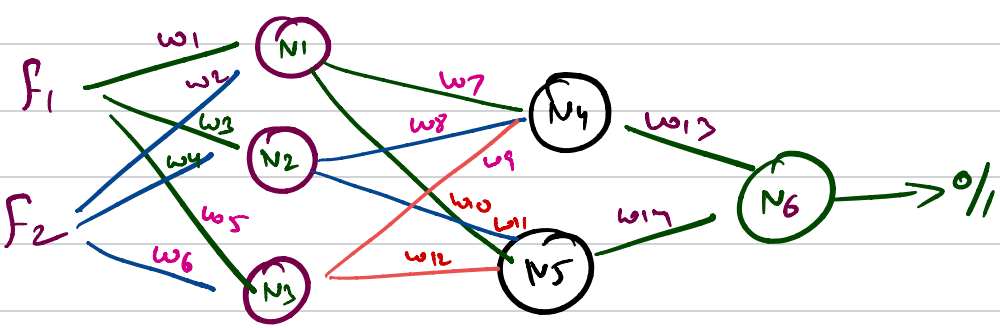
$b - 10 \rightarrow \sigma(10)$

$c - 15 \rightarrow \sigma(15)$

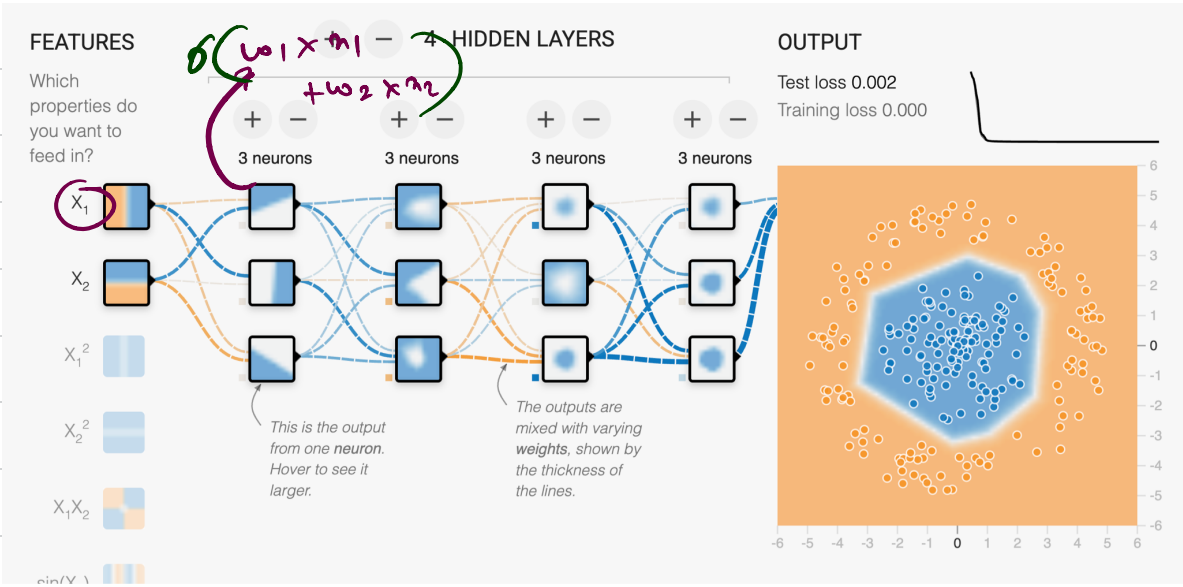
$c - b \neq b - a$

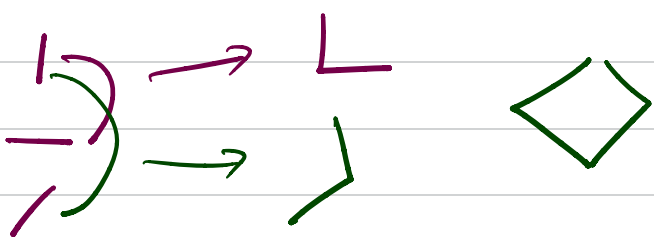
Why do you think, we need to combine multiple neutrons???


FEATURES
Which properties do you want to feed in?
6 HIDDEN LAYERS
6 neurons
7 neurons
7 neurons
5 neurons
5 neurons
5 neurons
X1
X2
X1²
X2²
X1X2
sin(X1)
sin(X2)
The outputs are mixed with varying weights, shown by the thickness of the lines.
OUTPUT
Test loss 0.000
Training loss 0.000
Colors shows data, neuron and weight values.
-1
0
1

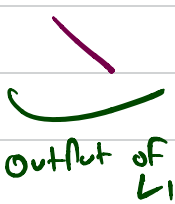

F1
F2
N1
N2
N3
N4
N5
N6
%
w1
w2
w3
w4
w5
w6
w7
w8
w9
w10
w11
w12
w13
w16

With deeper-layers, you get to learn more and more complicated non-linear combination of features.

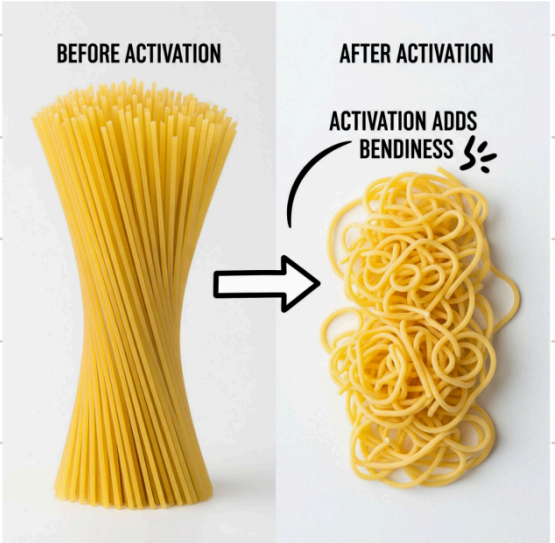


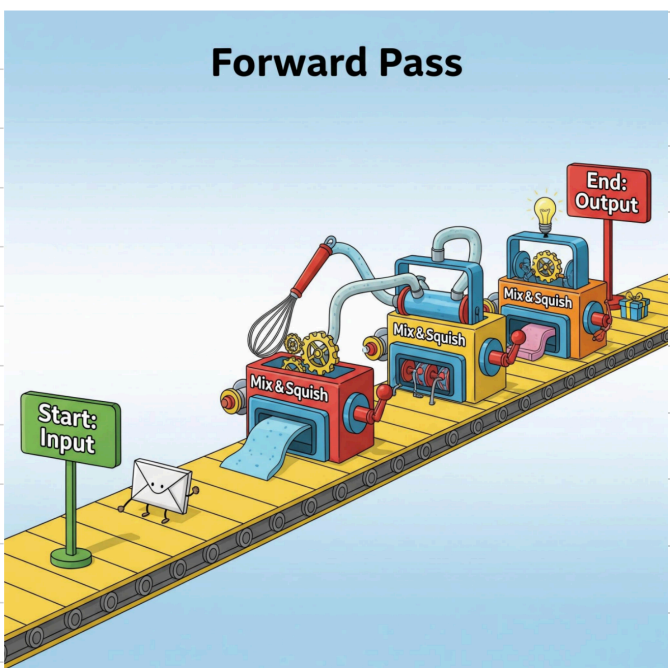


10 minutes.

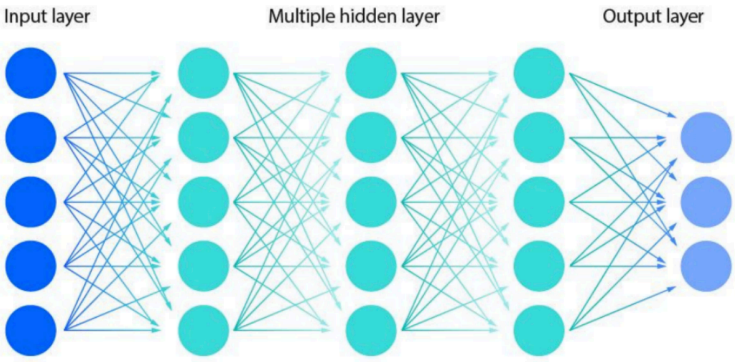


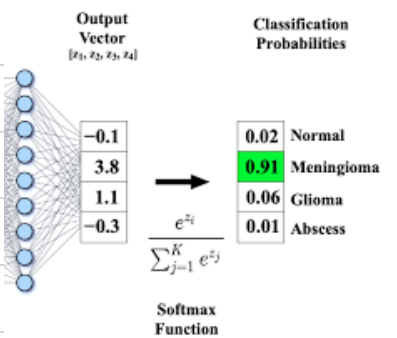
Vanishing And Exploding Gradients.


BEFORE ACTIVATION
AFTER ACTIVATION
ACTIVATION ADDS
BENDINESS


Forward Pass
Start:
Input
Mix & Squish
Mix & Squish
Mix & Squish
End:
Output

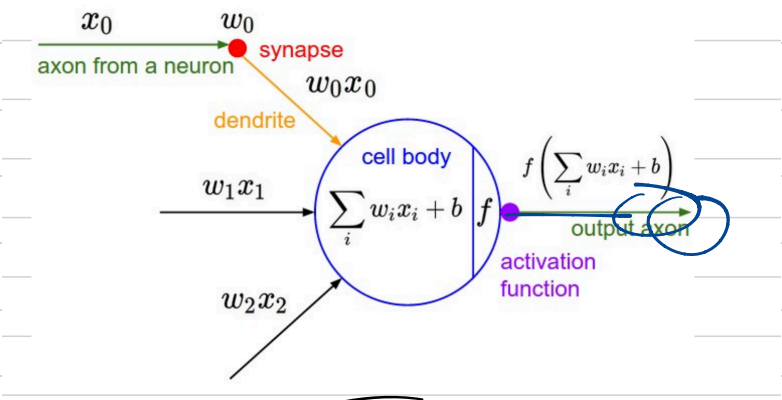
Deep neural network

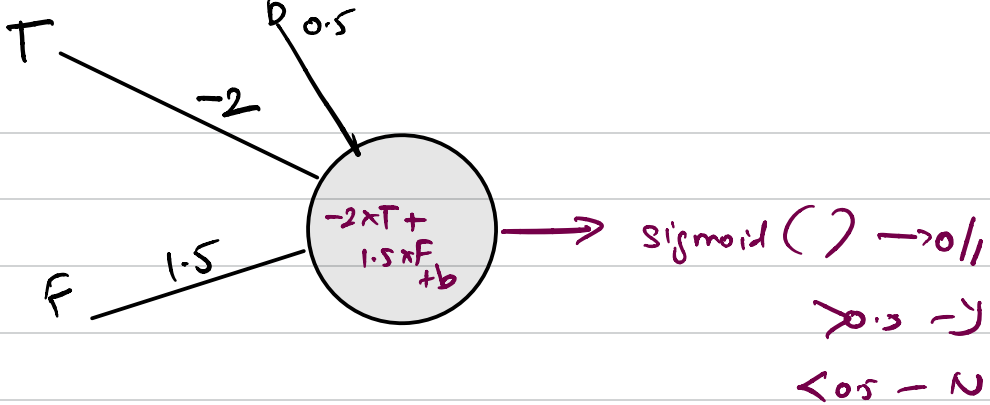




$$\frac{e^{z_i}}{\sum_j e^{z_j}}$$

$$\frac{e^{-0.1}}{e^{-0.1} + e^{3.8} + e^{1.1} + e^{-0.3}}$$





You combine log-red model, across multiple layers, and multiple model/ neuron in each layer, you get NN with sigmoid as activation.